Sarasij Maitra

Department of Mathematics and Statistics, Haverford College
370 Lancaster Avenue, Haverford, PA 19041
Personal Webpage
sarasij93@gmail.com

Curriculum Vitae

Education

2025–Present **Visiting Assistant Professor**, *Haverford College*

2022–2025 **Postdoctoral Researcher/Lecturer**, *University of Utah*

2016–2022 **PhD in Mathematics**, *University of Virginia*

2014–2016 **M.Math**, *Indian Statistical Institute*, Kolkata

2011–2014 **B.Math**, *Indian Statistical Institute*, Bangalore

Awards and Achievements

- **Postdoc Teaching Award in Mathematics, University of Utah, 2025**
- **AMS–Simons Travel Grant 2023–2025**
- The Joint Mathematics Meetings 2022 Travel Grant (Later shifted to a virtual venue)
- Jefferson Scholars Foundation Research Prize, 2021.
- **All-University Graduate Teaching Award, University of Virginia, 2019-2020**.
- (2019) Travel Grant for Mathematical Sciences Research Institute 883 Commutative Algebra and its Interaction with Algebraic Geometry.
- 1st Division with Distinction in M. Math and B. Math.

Mentoring Experience

2025-Present **Senior Thesis Advisor**, *Haverford College*,
Regular meetings for a year with students to discuss and develop their senior theses in areas such as Commutative Algebra, Machine Learning and Cryptography.
Current Students: Jose Salas Esquivel (Commutative Algebra), Avery Barnett (Cryptography), Jonah Levine (Deep Learning)

2024-2025 **Research Experiences For Undergraduates (REU) with William Zhang**, *Undergraduate, University of Utah*,
Publication: ***Trace does not preserve Reflexivity***,
Independent reading/research on Group Theory from *Abstract Algebra by Dummit & Foote* and applications of groups in Cryptography; Trace ideals in Numerical Semigroup Rings; sponsored by the Univ. of Utah Mathematics department with a student stipend of $1500-$2500; project joined by visitor Prof. H. Lindo around Spring 2025.

2022-2025 **AWM Mentor to Jacob Strong, Macy Dattile, Yuvraj Malik**, *Undergraduates, University of Utah*
Biweekly meetings to discuss various mathematical as well as career exploration topics.

Papers and Preprints

- **Optimal Parametrizations and Valuations** with R. Hübl, C. Huneke and V. Mukundan
- **Trace does not preserve Reflexivity** with H. Lindo and W. Zhang
- **Unstable elements in cohomology and a question of Lescot** with S. B. Iyengar and T. Tribone
- **A family of simplicial resolutions which are DG-algebras** with J. Cameron, T. Chau, T. Tribone
- **Annihilators of (co)homology and their influence on the trace Ideal** with J. Lyle (Submitted)
- **Two Criteria for Quasihomogeneity** with V. Mukundan; Proceedings of the American Mathematical Society 152.06 (2024): 2369-2375.
- **Extremal behavior of reduced type of one dimensional rings** with V. Mukundan (Submitted)
- **Partial Trace Ideals, Torsion and Canonical Module**; Journal of Algebra, 652: 1- 19, 2024.
- **Valuations and Nonzero Torsion in Module of Differentials** with V. Mukundan; Bulletin des Sciences Mathématiques, Volume 187, October 2023, 103287
- **Finding Points On Varieties with *MACAULAY2*** with S. Bisui, T. T. Nguyễn, Z. Jiang and K. Schwede; Journal of Software for Algebra and Geometry, Vol 13 (2023), 33-43
- **On Reflexive and I-Ulrich Modules over Curve Singularities**, with H. Dao and P. Sridhar; Trans. Amer. Math. Soc. Ser. B 10 (2023), 355-380
- **Traceable PRFs: Full Collusion Resistance and Active Security** with D. J. Wu; **PKC 2022**
- **Partial Trace Ideals and Berger's Conjecture**; Journal of Algebra, 598: 1–23, 2022.
- **Torsion in Differentials and Berger's Conjecture** with C. Huneke and V. Mukundan; Res Math Sci 8, 60 (2021)

Teaching

Fall 2025 **Visiting Assistant Professor**, *Haverford College*
MATH 121A04-05 Multivariable Calculus (Flipped), MATH 390: Commutative Algebra

Spring 2025 **Assistant Professor (Lecturer)**, *University of Utah*
MATH 3210-002 Foundations of Real Analysis I

Fall 2024 **Assistant Professor (Lecturer)**, *University of Utah*
MATH 2270-001,006 Linear Algebra

Spring 2024 **Assistant Professor (Lecturer)**, *University of Utah*
MATH 2270-001 Linear Algebra

Fall 2023 **Assistant Professor (Lecturer)**, *University of Utah*
MATH 1090-003, MATH 1090-006 Business Algebra

Spring 2023 **Assistant Professor (Lecturer)**, *University of Utah*
MATH 3150-004 PDE's for Engineers

Fall 2022 **Assistant Professor (Lecturer)**, *University of Utah*
MATH 1321-001 Accelerated Engineering Calculus II

Spring 2022 **Graduate Teaching Instructor**, *University of Virginia*
MATH 1310-400 Calculus I (Flipped Model)

Fall 2021 **Graduate Teaching Instructor**, *University of Virginia*
MATH 1410-001 Financial Mathematics

Fall 2020 **Graduate Teaching Instructor**, *University of Virginia*
MATH 1320-200 Calculus II (Online Flipped Model)

Fall 2019 **Graduate Teaching Instructor**, *University of Virginia*
MATH 1310-001 Calculus I (Flipped Model)

Spring 2019 **Graduate Teaching Instructor**, *University of Virginia*
MATH 1220-009 A Survey of Calculus II

Spring 2018 **Graduate Teaching Instructor**, *University of Virginia*
MATH 1210-008 A Survey of Calculus I

Fall 2017 **Graduate Teaching Instructor**, *University of Virginia*
MATH 1210-007 A Survey of Calculus I

Spring 2017 **Graduate Teaching Assistant**, *University of Virginia*
MATH 1310-100 Calculus I, MATH 3250-300 Ordinary Differential Equations

Fall 2016 **Graduate Teaching Assistant**, *University of Virginia*
MATH 1310-500, MATH 1310-700 Calculus I

Conferences and Seminars Organized

April 2025 **MATH FOR ALL in Salt Lake City, Utah**, University of Utah, Co-organizers: T. Tribone, K. Vinhage

Spring 2024 **Commutative Algebra Seminar**, University of Utah, Co-organizers: Eamon Quinlan-Gallego, Karl Schwede

April 2024 **MATH FOR ALL in Salt Lake City, Utah**, University of Utah, Co-organizers: P. Mcdonald, T. Tribone

2023 **University of Utah MathCircle**, University of Utah, Co-organizers: A. Borisyuk, K. Vinhage, X. Shen

October 2022 **AMS Special Session on Building Bridges Between Commutative Algebra and Nearby Areas**, *Fall Western Sectional Meeting*, University of Utah, Co-organizers: J. Cameron, T. Tribone

Service

- Reviewer for various mathematics journals
- AWM Mentor at the University of Utah, 2022-2025

- High School Sectional Leader, Mathcircle University of Utah, Spring and Fall 2023
- Graduate Teaching Mentor at the University of Virginia, 2020-2021

Software

- *RandomPoints*, (Macaulay 2), with S. Bisui, T. T. Nguyễn, Z. Jiang, K. Schwede.
- *SwitchingFields*, (Macaulay 2), with Z. Jiang.

Invited Talks and Poster Presentations

July 2025 **Discussions on Berger's Conjecture**, *Recent Developments in Commutative Algebra*, Mathematical Congress of the Americas 2025, Miami, FL

October 2024 **On Nearly Gorenstein Rings and a Generalization of a Conjecture of Tachikawa**, *Special Session on Singularities in commutative algebra*, 2024 Fall Eastern Sectional Meeting, University at Albany, Albany, NY

October 2024 **On Nearly Gorenstein Rings and a Generalization of a Conjecture of Tachikawa**, *Special Session on Topics in Commutative Algebra and Algebraic Geometry*, 2024 Fall Southeastern Sectional Meeting, Georgia Southern University, Savannah, GA

September 2024 **Reduced type of one dimensional complete local domains**, *AMS Special Session on Commutative algebra and connections to combinatorics*, 2024 Fall Central Sectional Meeting, University of Texas – San Antonio

January 2024 **Extremal Behaviour of Reduced Type**, *Recent Developments in Commutative Algebra*, Joint Mathematics Meetings, San Francisco, CA

April 2023 **On the reduced type of one dimensional analytic k-algebras**
Morgantown Algebra Days (MAD), West Virginia University

April 2023 **The Frobenius Problem**
MathCircle, Salt Lake City, University of Utah

February 2023 **Frobenius Problem and Numerical Semigroup Rings**
Math for All, Salt Lake City, University of Utah

Februrary 2023 **Module of Differentials and Berger's Conjecture**
Algebra Seminar, University of Georgia

October 2022 **Discussions on Berger's Conjecture**
Commutative Algebra Seminar, University of Utah

April 2022 **Notes On Berger's Conjecture**
Graduate Seminar, University of Virginia

April 2022 **On Reflexive and I-Ulrich Modules**, *AMS Special Session on Commutative Algebra*, Joint Mathematics Meetings

March 2022 **Torsion in Module of Differentials**, *Special Session on Recent Developments in Commutative Algebra*, AMS Spring Central Sectional Meeting at Purdue University, West Lafayette, IN

March 2022 **Traceable PRFs: Full Collusion Resistance and Active Security**
PKC, Japan (virtual)

January 2022	**Partial Trace Ideals and Berger's Conjecture** Virtual Commutative Algebra Seminar, University of Illinois at Chicago
April 2021	**Discussions on I-Ulrich Modules** Virtual Commutative Algebra Seminar, Purdue University
April 2021	**An Introduction To Gröbner Bases**, *(Joint Talk with Stephanie Shand)* Graduate Seminar, University of Virginia
February 2021	**Discussions on I-Ulrich Modules** Algebra Seminar, West Virginia University
January 2021	**Discussions on Berger's Conjecture** Joint Mathematics Meetings
October 2020	**A Study of Colength in Dimension One** Commutative and Homological Algebra Market Presentations (CHAMP)
June 2020	**Poster: An Approach to Berger's Conjecture** Early Commutative Algebra Researchers (eCARs)
June 2020	**A Simple Study of Colength** Graduate Student Seminar, New Mexico State University
March 2020	**"Berger's Conjecture from the Viewpoint of an Invariant of the Module of Differentials"–An Approach to Berger's Conjecture** A Zoom Session on Commutative Algebra

Memberships

- Association for Women in Mathematics (AWM)
- American Mathematical Society (AMS)

Other Projects

Summer 2024	Deep learning project on Detecting Skin Cancer using Image Processing – The Erdös Institute
Summer 2024	Data Science project on Predicting house prices in King County, WA – The Erdös Institute [LINK]
Jan - May, 2016	Project on *Bargaining Games with Time-Independent and Time-Dependent Preferences* under Prof. Souvik Roy of Indian Statistical Institute, Kolkata. [Project Report]
Dec, 2014	Reading Project on *Hopf Algebra* under Prof. Jyotishman Bhowmick of Indian Statistical Institute, Kolkata.
May - July, 2013	Indian Academy of Sciences Summer Project on *Algebraic Geometry* under Prof. Kapil Hari Paranjape at Indian Institute of Science Education and Research, Mohali. [Project Report]

Skills

Language Skills	English, Bengali, Hindi
Technologies	Macaulay 2, LaTeX, MATLAB, Python, Jupyter Notebooks